MILITARY SPECIFICATION SHEET

METER, TIME TOTALIZING, MINIATURE, DIGITAL,
28 VDC OR 26 VAC, 50 TO 2400 HZ, SQUARE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the meter described herein shall consist of this specification and the latest issue in effect of MIL-M-7793.

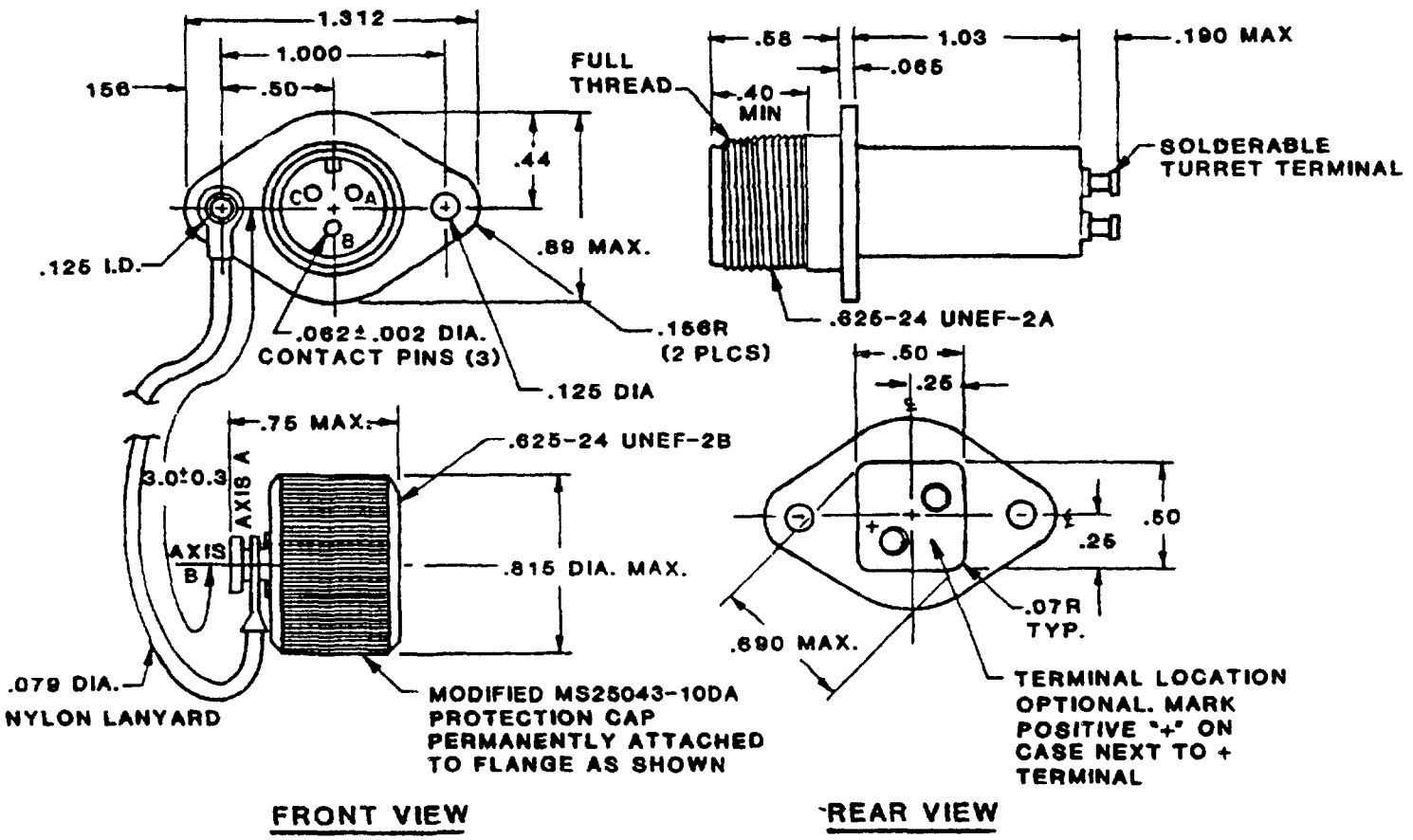


FIGURE 1.

AMSC N/A FSC 6645

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REQUIREMENTS:

1. Dimensions are in inches. Unless otherwise specified, tolerances are ±.02 for two-place decimals and ±.015 for three-place decimals.

2. The mounting flange, threaded extension and threaded protection cap shall be constructed of aluminum alloy with an olive drab chromate coating over cadmium plating.

3. The protection cap shall be in accordance with MS25043-10DA except for the finish and means of attachment. The finish shall be in accordance with Requirement 2 and the metallic chain shall be replaced with a nylon lanyard.

Performance:

Dielectric strength: The meter shall be capable of withstanding the application of 600 VRMS (room) and 350 VRMS (altitude) between the terminals and the case.

Life accuracy: ±0.1% from -65° to +125°C and 10 to 34 VDC, or 20 to 30 VAC at 50 to 2400 Hz. Qualification tests shall be performed on 3 meters each, for DC and AC accuracy.

Power consumption: Power consumption at 28 VDC shall not exceed 50 milliwatts and at 26 VAC, 400 Hz AC shall not exceed 25 milliwatts.

Ripple voltage: When operated on DC, the meter shall operate normally when subjected to the ripple requirements of 5.2.2, 5.2.2.1 and Figure 7 of MIL-STD-704A.

Operating temperature range: -65° to +125°C.

Part number: M7793/14-1.

Time range: Total readout of the meters shall be 99999.99 hours.

Meter output: The output from the meter shall be in a serial binary coded decimal format. The output impedance shall be 100K ± 10%. A logical "zero" shall be between 0.0 and +0.2 volts. A logical "one" shall be between +3.3 and +6.6 volts. The output pin assignments are shown in Figure 2 and the output data format is shown in Figure 3.

Transient protection: When operated on DC, no temporary or permanent degradation shall be produced in the meter for the input voltage and time values shown in MIL-STD-704A, Figure 17 and Figure 9, Curve 1 (600 V and 80 V transients, respectively). When operated on AC, no temporary or permanent degradation or other malfunction shall be produced in the time totalizing meter if the input voltage should increase to 41 VRMS at 320 to 480 Hz for a period of 150 milliseconds maximum.

Durability of cap and threads: No mechanical or electrical degradation after 100 cycles in accordance with MIL-C-5015.

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Voltage range: Operating voltage shall be from 10 to 34 VDC or 20 to 30 VAC, 50 to 2400 Hz.

Durability of connectors and contacts: No mechanical or electrical degradation after 100 cycles in accordance with MIL-C-5015. Operational accuracy shall be checked at the end of the test.

Weight: Weight shall be not greater than 1.0 ounce with the attached protection cap.

Tensile strength of lanyard: The lanyard shall withstand a tensile load of 25 pounds when applied to Axis "A" and Axis "B" as shown on Figure 1.

Qualification tests: Qualification tests will be conducted in accordance with MIL-M-7793, except that the voltage, temperature and accuracy shall be as specified herein. Salt spray, moisture resistance, vibration, shock, electromagnetic compatibility and low temperature storage shall be conducted with the protective cap in place. Hermetic seal test is not applicable.

NOTES:

1. Elapsed time is read from the M7793/14-1 meter by mating the M7793/14-1 meter's connector with the M7793/12-1 reader's connector and operating the reader.

2. Digital readout and meter face plate per MIL-M-7793 are not applicable since the digital readout is provided by the separate M7793/12-1 reader.

3. AC operation may be from either sinusoidal or square wave sources.

4. The meter is designed to have an electrical ground and a continuous EMI/RFI shield with the threaded protection cap in place.

5. Contacts shall meet all the requirements of silver plated solder contacts from MIL-C-5015.

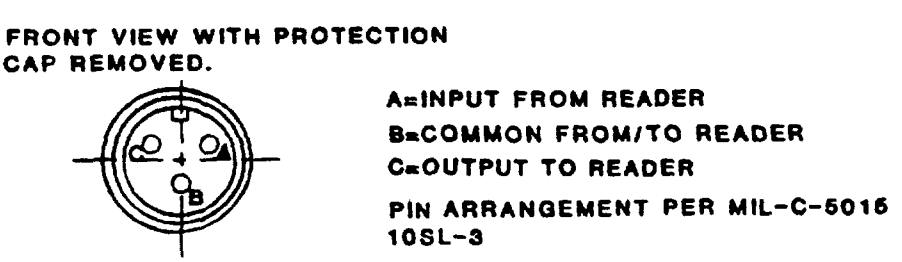


FIGURE 2. Readout pin assignment.

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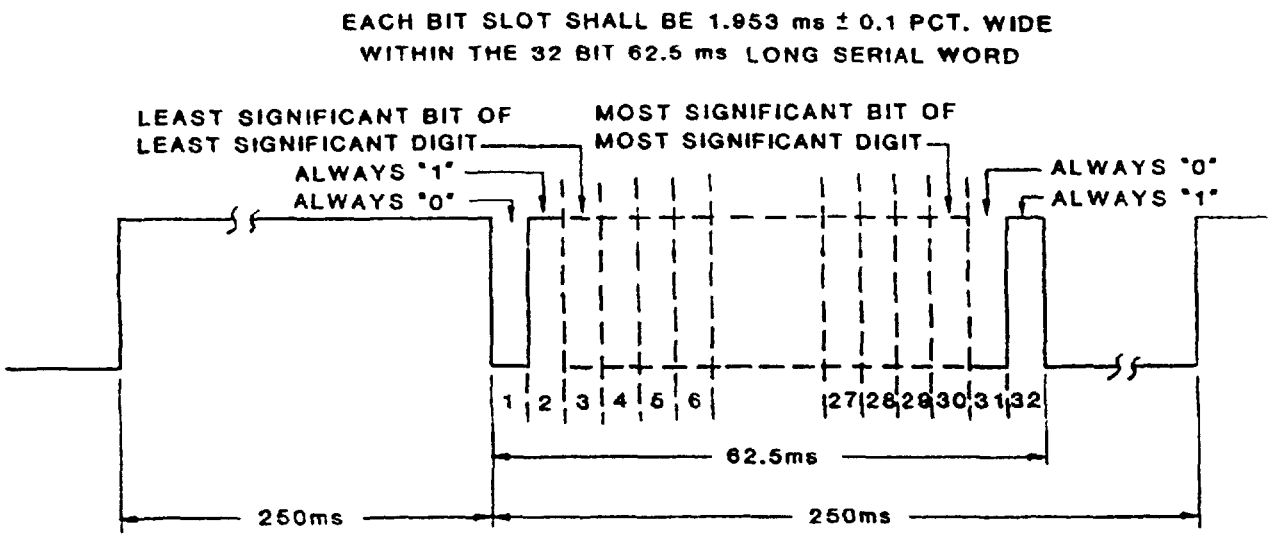


FIGURE 3. Output data format.

Custodians:
Navy - AS
Army -
Air Force - 99

Users:
Navy - SH, OH

Reviewer:
Army - MI

Preparing activity:
Navy - AS
(Project Number 6645-401)

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